

4.2 Cell transport mechanisms

- i Know the structure of the cell surface membrane with reference to the fluid mosaic model.
- ii Understand how passive transport is brought about by:
 - diffusion
 - facilitated diffusion (through carrier proteins and protein channels)
 - osmosis.
- iii Understand how the properties of molecules affects how they are transported, including solubility, size and charge.
- iv Know that large molecules can be transported into and out of cells through the formation of vesicles, in the processes of endocytosis and exocytosis.

CORE PRACTICAL 5: Investigate the effect of temperature on beetroot membrane permeability.

CORE PRACTICAL 6: Determine the water potential of plant cells.

Water potential = turgor pressure + osmotic potential

$$\psi = P + \pi$$